CHAPTER 1

Some Basic Concepts of Chemistry

Significant Figures, Laws of Chemical Combinations

1. The density of the solution is 2.15 g mL^{-1}, then mass of 2.5 mL solution in correct significant figures is: *(2022 Re)*

a. 53.75 g b. 5375×10^{-3} g
c. 5.4 g d. 5.38 g

2. Equal masses of H_2, O_2 and methane have been taken in a container of volume *V* at temperature 27°C in identical conditions. The ratio of the volumes of gases $H_2 : O_2$: methane would be: *(2014)*

a. 8 : 16 : 1 b. 16 : 8 : 1
c. 16 : 1 : 2 d. 8 : 1 : 2

Atomic and Molecular Masses

3. Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2. When 0.1 mole of XY_2 weighs 10 g and 0.05 mole of X_3Y_2 weighs 9 g, the atomic weights of X and Y are: *(2016)*

a. 20, 30 b. 30, 20 c. 40, 30 d. 60, 40

Mole Concept and Molar Masses

4. 1.0 g of H_2 has same number of molecules as in: *(2024 Re)*

a. 14 g of N_2 b. 18 g of H_2O
c. 16 g of CO d. 28 g of N_2

5. The highest number of helium atoms is in *(2024)*

a. 4 g of helium b. 2.271098 L of helium at STP
c. 4 mol of helium d. 4 u of helium

6. Which one of the followings has **maximum** number of atoms? *(2020)*

a. 1 g of Mg(s) [Atomic mass of Mg = 24]
b. 1 g of O_2(g) [Atomic mass of O = 16]
c. 1 g of Li(s) [Atomic mass of Li = 7]
d. 1 g of Ag(s) [Atomic mass of Ag = 108]

7. One mole of carbon atom weighs 12g, the number of atoms in it is equal to. *(2020-Covid)*

(Mass of carbon- 12 is 1.9926×10^{-23} g)

a. 6.022×10^{22} b. 12×10^{22}
c. 6.022×10^{23} d. 12×10^{23}

8. In which case is number of molecules of water maximum?

a. 18 mL of water *(2018)*
b. 0.18 g of water
c. 10^{-3} mol of water
d. 0.00224 L of water vapours at 1 atm and 273 K

9. A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture? *(2015)*

a. 16 : 1 b. 2 : 1 c. 1 : 4 d. 4 : 1

10. The number of water molecules is maximum in: *(2015 Re)*

a. 18 moles of water b. 18 molecules of water
c. 1.8 gram of water d. 18 gram of water

11. If Avogadro number N_A, is changed from 6.022×10^{23} mol^{-1} to 6.022×10^{20} mol^{-1}, this would change: *(2015 Re)*

a. The ratio of elements to each other in a compound
b. The definition of mass in units of grams
c. The mass of one mole of carbon
d. The ratio of chemical species to each other in a balanced equation

Percentage Composition, Empirical & Molecular Formula

12. A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is:
(Given atomic masses of A = 64; B = 40; C = 32u) *(2024)*

a. AB_2C_2 b. ABC_4 c. A_2BC_2 d. ABC_3

13. An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is: [Atomic wt. of C is 12, H is 1] *(2021)*

a. CH_2 b. CH_3 c. CH_4 d. CH

Stoichiometry and Stoichiometric Calculations

14. 1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to *(2024)*

a. Zero mg b. 200 mg c. 750 mg d. 250 mg

15. The right option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40)

$\left[CaCO_3 \xrightarrow{1200\,K} CaO + CO_2\right]$ *(2023)*

a. 1.32 g b. 1.12 g c. 1.76 g d. 2.64 g

16. The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is: *(2019)*

a. 10 b. 20 c. 30 d. 40

17. A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with concentration H_2SO_4. The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be: *(2018)*

a. 1.4 b. 3.0 c. 4.4 d. 2.8

18. What is the mass of the precipitate formed when 50 mL of 16.9% solution of $AgNO_3$ is mixed with 50 mL of 5.8% NaCl solution? *(2015 Re)*

(Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)

a. 3.5 g b. 7 g c. 14 g d. 28 g

19. 20.0 g of a magnesium carbonate sample decomposes on heating to give carbon dioxide and 8.0 g magnesium oxide. What will be the percentage purity of magnesium carbonate in the sample? *(2015 Re)*

(Atomic weight of Mg = 24)

a. 96 b. 60 c. 84 d. 75

20. When 22.4 litres of H_2(g) is mixed with 11.2 litres of Cl_2(g), each at STP, the moles of HCl(g) formed is equal to: *(2014)*

a. 2 mol of HCl(g) b. 0.5 mol of HCl(g)

c. 1.5 mol of HCl(g) d. 1 mol of HCl(g)

21. 1.0 g of magnesium is burnt with 0.56 g O_2 in a closed vessel. Which reactant is left in excess and how much? *(2014)*

(Atomic weight Mg = 24; O = 16)

a. O_2, 0.16 g b. Mg, 0.44 g

c. O_2, 0.28 g d. Mg, 0.16 g

Concentration Terms

22. The density of 1 M solution of a compound 'X' is 1.25g mL^{-1}. The **correct** option for the molality of solution is (Molar mass of compound X = 85 g): *(2023-Manipur)*

a. 0.705 m b. 1.208 m c. 1.165 m d. 0.858 m

23. What mass of 95% pure $CaCO_3$ will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction? *(2022)*

$CaCO_{3(s)} + 2HCl_{(aq)} \longrightarrow CaCl_{2(aq)} + CO_{2(g)} + 2H_2O_{(l)}$

[Calculate upto second place of decimal point]

a. 9.50 g b. 1.25 g c. 1.32 g d. 3.66 g

24. 6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of solution is: *(2013)*

a. 0.02 M b. 0.01 M

c. 0.001 M d. 0.1 M

25. An excess of $AgNO_3$ is added to 100 mL of a 0.01 M solution of dichlorotetraaquachromium(III) chloride. The number of moles of AgCl precipitated would be: *(2013)*

a. 0.001 b. 0.002 c. 0.003 d. 0.01

26. How many grams of concentrated nitric acid solution should be used to prepare 250 mL of 2.0 M HNO_3? The concentrated acid is 70 % HNO_3: *(2013)*

a. 45.0 g conc. HNO_3 b. 90.0 g conc. HNO_3

c. 70.0 g conc. HNO_3 d. 54.0 g conc. HNO_3

Answer Key

1. (c)	**2.** (c)	**3.** (c)	**4.** (a)	**5.** (c)	**6.** (c)	**7.** (c)	**8.** (a)	**9.** (d)	**10.** (a)
11. (c)	**12.** (d)	**13.** (b)	**14.** (d)	**15.** (c)	**16.** (c)	**17.** (d)	**18.** (b)	**19.** (c)	**20.** (d)
21. (d)	**22.** (d)	**23.** (c)	**24.** (b)	**25.** (a)	**26.** (a)				